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A Activation Decomposition with SNMF

In this section we describe in detail the process used to decompose MLP activations with `SNMF`.

Initialization The non-negative activation matrix $Y \in \mathbb{R}_{\geq 0}^{k \times n}$ is sampled element-wise from a uniform distribution on $[0, 1]$, and the feature matrix $Z \in \mathbb{R}^{d \times k}$ is sampled from a standard normal distribution. We chose this initialization for simplicity and left optimizations for future work.

Feature update With Y fixed we obtain Z:

$$Z \leftarrow AY^\top \big(YY^\top + \lambda I\big)^{-1},$$

where $\lambda > 0$ is a small regularisation constant that prevents ill-conditioning.

Activation update Keeping Z fixed, the non-negative activation matrix Y is updated using the multiplicative rule derived from the `SNMF` objective:

$$Y \leftarrow Y \odot \sqrt{\frac{\big[Z^\top A\big]_+ + \big[Z^\top Z\big]_- Y}{\big[Z^\top A\big]_- + \big[Z^\top Z\big]_+ Y}},$$

where $[X]_+$ ($[X]_-$) denotes the element-wise positive (negative) part of X, and $\odot$ is the Hadamard product. This update preserves the non-negativity of Y.

Sparsity and Normalization We impose an ℓ_0 constraint during training by applying a hard winner-take-all (WTA) operator to every column of Z, keeping only the largest $p\%$ of entries (by absolute value) and zeroing the remainder. In our experiments we used $p = 1$. To stabilize training we normalize every column of Y to unit ℓ_2 norm and rescale the corresponding column of Z so that ZY is unchanged. The sparsity method is adapted from Peharz and Pernkopf (2012), who employed a similar approach for enforcing sparsity in Non-negative Matrix Factorization.

B Cosine Similarity versus Projection

If we consider the formula for projection we get,

$$\begin{aligned} m_{\text{proj}}(S) &= \frac{1}{|S|} \sum_{\mathbf{x} \in S} \mathbf{f}^\top \mathbf{x} \\ &= \|\mathbf{f}\| \cdot \frac{1}{|S|} \sum_{\mathbf{x} \in S} \|\mathbf{x}\| \cos \theta_x \end{aligned} \tag{7}$$